

Parallel and HPC HW5 Report

Link to github repository:

<https://github.com/DMelkonyan111/Parallel-and-HPC-Homeworks>

Assignment 1

Assignment 1 involved developing a parallel log classifier using OpenMP. To begin, I generated an array of 20,000 logs each comprising a request_id, user_id, and response_time_ms values. Inside a parallel region consisting of four threads, I made use of one single construct for initializing the array of logs. In particular, each log was initialized with an incrementing request_id value, a user_id based on the identity of the thread handling it, and a random response_time_ms value. Following initialization, I utilized #pragma omp for for assigning each log a category (either FAST <100 ms, MEDIUM 100-300 ms, or SLOW >300 ms). While doing this, I allowed each thread to maintain their own counts (i.e., counters) in order to ensure there was no race condition. Eventually, after classification, one last single construct helped aggregate the counts from the individual threads and output the totals.

```
davitm@ubuntu-aia-os:~/workshop/Homeworks/5$ ./Ex1
Number of fast logs: 3939
Number of medium logs: 7977
Number of slow logs: 8083
```

Assignment 2

Assignment 1 consisted in creating a parallel log classifier using OpenMP. First, I created an array of 20,000 logs, where every log had request_id, user_id, and response_time_ms values. Within a parallel section that had four threads, I used a single statement to initialize the array of logs. Specifically, every log was initialized by its request_id incremented value, user_id depending on the thread executing it, and randomly assigned response_time_ms value. After that, I employed the #pragma omp for statement to classify every log into either fast (<100 ms), medium (100-300 ms), or slow (>300 ms) categories. During this process, I let each thread use its own counter so that there would not be any race conditions. Finally, after classifying all the logs, the final count results were obtained using another single statement.

```
davitm@ubuntu-aia-os:~/workshop/Homeworks/5$ ./Ex2
All priorities have been assigned
The number of high priority orders is 3895
```